

JUNSU KIM

Curriculum Vitae

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Research Interests

- Egocentric and embodied AI for tool use, physical interactions, and action-driven state changes.
 - 3D/4D scene understanding and action-centric world models trained on human and robot data.
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Education

Korea Advanced Institute of Science and Technology (KAIST)

B.S. in Electrical Engineering

GPA: 4.09/4.3 (Major GPA: 4.19/4.3)

Military Service, Republic of Korea Army: Jul 2022 – Jan 2024

Daejeon, South Korea

Mar 2021 – *Expected Aug 2027*

Nanyang Technological University (NTU)

Exchange Student

School of Electrical and Electronic Engineering

Singapore

Jan 2026 – May 2026

Publications

[1] **EgoTools: Towards Tool-Centric Reasoning in Real-World Egocentric Videos.**

[\[Project Page\]](#) [\[Paper\]](#).

Shulin Tian*, **Junsu Kim***, Shuai Liu*, Hao Li*, ..., Ziwei Liu.

EMNLP 2026 (Main Conference).

[2] **C3G: Learning Compact 3D Representations with 2K Gaussians.**

[\[Project Page\]](#) [\[Paper\]](#).

Honggyu An*, Jaewoo Jung*, ..., **Junsu Kim**, ..., Seungryong Kim.

CVPR 2026.

[3] **DreamX: Video Diffusion Model Is A Universal Latent Action Learner.**

Jisoo Kim*, Chaehyun Kim*, **Junsu Kim**, ..., Seungryong Kim.

Under review at ICLR 2027.

*Equal contribution.

Research Experience

MAUM AI, WoRV Division, Future Intelligence Lab

Aug 2026 – Present

Research Intern

- Evaluating real-world vision-language-action and world-action models and developing human-aware embodied evaluation tasks, with a focus on physical reasoning and action-conditioned behaviors.

S-Lab (MMLab), NTU

Jan 2026 – Present

Research Intern (Advisor: [Prof. Ziwei Liu](#), NTU)

- Co-led **EgoTools**, a real-world egocentric dataset and benchmark for tool-centric reasoning, built from **100+ hours** of long-horizon tool-use videos.
- Designed annotation and QA workflows for physical-state, procedural, affordance, and tool-use reasoning; coordinated data collection and quality control; and co-led construction of a **1K-QA** benchmark.

- Fine-tuned **Qwen3-VL-8B** on **220K** multimodal training examples, improving EgoTools-Bench accuracy from **50.0%** to **60.9%**; conducted cross-benchmark generalization, failure-case, and catastrophic-forgetting analyses.

Computer Vision Lab (CVLab), KAIST AI

Jun 2025 – Present

Research Intern (Advisor: [Prof. Seungryong Kim](#), KAIST AI)

- *Jun 2026 – Oct 2026*: Designed and conducted experiments for **DreamX**, evaluating latent-action representation quality, cross-embodiment video/action transfer, and representation behavior through baseline studies and targeted experiments motivated by qualitative observations.
- *Jun 2025 – Nov 2025*: Contributed to **C3G**, investigating feed-forward 3D reconstruction and novel-view synthesis with Gaussian Splatting, VGGT, and NoPoSplat, with a focus on geometry quality and representation behavior.

Video and Image Computing Lab (VICLab), KAIST

Dec 2024 – Jun 2025

Research Intern (Advisor: [Prof. Mun Churl Kim](#), KAIST EE)

- Implemented and evaluated object-detection pipelines spanning two-stage, YOLO-style, and Vision Transformer architectures for few-shot and anomaly detection tasks.

Honors & Scholarships

KAIST Presidential Fellowship (15th)

Feb 2025 – Present

National Science & Engineering Scholarship for Academic Excellence

Aug 2025 – Present

Woon Hae Scholarship (12th)

Feb 2025 – Dec 2025

KAIST Dean's List, College of Engineering

Fall 2021, Spring 2024

awarded to the top 3% of students

KAIST Shared Lives Award (Top-ranked Recipient)

Feb 2026

For exceptional and sustained blood donation service

Leadership & Service

President, EERun (KAIST EE Running Club)

Sep 2024 – Dec 2025

Organized weekly training sessions and semester events for KAIST EE students.

Google Student Ambassador

Sep 2025 – Dec 2025

Promoted Gemini Pro to KAIST students through short-form video content.

Additional Activities

2022 – Present

KAIST AI Studying Club, Young Engineers Honor Society (YEHS), SilverLining, Freshman Proctor.

Skills

Programming

C/C++, Python, PyTorch, Git, LaTeX

Research Tools

video preprocessing, dataset curation, annotation pipelines, benchmark construction, evaluation scripts

Languages

Korean (native), English (fluent, iBT TOEFL 110)